Abhishek Karna

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Education

- **Duke University** — Durham, USA
  BS Physics and Mathematics, Cert. Philosophy, Politics, and Economics (PPE); GPA-3.87 — *Aug. 2022 - May 2026*

Skills Summary

- **Selected Courses**: Quantum mechanics-I&II, Representation theory*, Quantum lattice Monte Carlo, Frustrated quantum spin system, Quantum materials-I&II, Complex analysis, Novel quantum magnets-I&II, General relativity*, Algebraic structures-I*, Real analysis*, Differential geometry. (*=Graduate course)
- **Research**: Numerical modeling of Hamiltonians, Quantum Monte Carlo, Exact diagonalization, X-ray diffraction, Laue scattering, Single crystal growth, Climate data modeling, HYSPLIT trajectory analysis, Data extraction, cleaning, and statistical inference.
- **Software**: Python (numpy, scipy, pandas, etc.), HPC Cluster, Version control, AI Models, LaTeX, OriginPro, FullProf

Research Experience

- **Frustrated Quantum Spin System** — Durham, NC
  Supervisor: Prof. Shailesh Chandrasekharan, Duke University — *May. 2024 - Present*
  - **Theoretical work**: Analyzed the established ground state derivation for a certain parameter range of the Shastry-Sutherland model (SSM), a geometrically frustrated 2D Heisenberg anti-ferromagnet; Applied the theory of angular momentum and addition of spins to a 16-spin lattice to perform a dimensionality reduction from 2^{16} to 12870 for a numerical ground state search in any parameter range; Formulated symmetry arguments to calculate total spin s for each energy state to recover the complete 2^{16} states; For Quantum Monte Carlo (QMC), performed the trotterrization of the Boltzmann factor in the partition function; Mapped the energy search problem as a sampling problem through ideas from path integral formulation of QM; Performed analytical calculations following rules of ergodicity and the principle of detailed balance from stochastic process theory to understand the constraints on the SSM Hamiltonian for algorithm development.
  - **Numerical modeling**: Devised a first principles framework to execute Exact Diagonalization (ED) for a $16-$spin lattice SSM Hamiltonian; performed a ground state search for different coupling ratios and calculated the S^2 matrices and total spin through diagonalization of all irreps; Developed the new *Projected Density Matrix Sampling (PDMS)* method and solved 2-, 3-, 4-, 16-, 64, 96-site systems accurately; currently investigating the algebraic lattice symmetries, "sign problem" in new projection subspaces, and multi-level ergodic sampling to improve the method's applicability. Paper* Presentation (ED)* Poster* Poster 2*
- **Quantum Materials Synthesis and Characterization** — Durham, NC
  Supervisor: Prof. Sara Haravifard, Duke University — *Aug. 2022 - Present*
  - **Synthesis techniques**: Conducted solid-state synthesis to make batches of different concentrations of Cr-doped $ZnAl_2O_4$, Lithium-doped ZnV_2O_4 (spinel compounds), and $NdFeO_3$ (topological magnon candidate); performed arc melting method for TmB_4 and HoB_4 (metallic SSM altermagnet candidate); synthesized single crystals of $NdFeO_3$ using optical floating zone.
  - **X-ray diffraction, Laue scattering, and PPMS**: Executed high-resolution X-ray diffraction with Rietveld refinement using the FullProf Suite and Origin, interpreting CIF and PCR files for obtaining the lattice parameters for structural characterization and general phase identification for purity checks; investigated the Vegard's law for structural deviations for doping experiments of spinels; conducted Laue scattering on single crystals for orientation alignment and magnetometric measurements using Quantum Design PPMS. Poster 1* Poster 2* Poster 3*
- **Atmospheric Pollution and Climate Physics** — Durham, NC
  Supervisor: Prof. Brian G. McAdoo, Duke University — *Nov. 2022 - Present*
  - **$PM_{2.5}$ modeling and Synoptic-scale Dynamics**: Applied air parcel backtrajectory modeling (HYSPLIT), aerosol optical depth (AOD) analysis, & potential source contribution functions (PSCF) integrating variety of meteorological and on-site pollution data to model transboundary $PM_{2.5}$ movement in Nepal and the Indo-Gangetic Plain (IGP) region, identifying pollutant transport pathways and cross-border influence. Presentation*
  - **Local pollution dynamics**: Examined diurnal $PM_{2.5}$ patterns, planet boundary layer height (PBLH), and ventilation coefficients (VC) to isolate local and non-local pollution sources, using moving average subtraction method for non-local effect quantification.
  - **Meteorological analysis**: Assisted in international field work projects by PhD students in the group; created heatwave maps for SE US. Currently developing Bayesian Networks method for Systems Dynamics-type analysis of climate-health questions.
- **Collider Physics for HL-LHC Upgrade** — Durham, NC
  Supervisor: Prof. Ashutosh Kotwal, Duke University — *Apr. 2023 - Aug. 2023*
  - **Wedge generation algorithm**: Designed and implemented an algorithm to segment into wedges the transverse and longitudinal sections of the concentric collider detectors for detecting DM-particle candidates as per SUSY theory.
  - **Spacepoint divider algorithm**: Devised an algorithm that takes HL-LHC collision spacepoint data, and divides them into the geometrically generated wedges for parallelized silicon-based track trigger particle track reconstruction for theorized metastable particles with extremely fast computational speed. Poster* Acknowledged for contributions in a published paper*
- **National Nanotechnology Research Laboratory, Tribhuvan University** — Kathmandu, Nepal
  Supervisor: Prof. Rameshwar Adhikari, Tribhuvan University — *Nov. 2021 - Aug. 2022*
  - **Sample preparation and investigation**: Prepared thin-oil films and conducted the UV-Visible Spectroscopy by operating Shimadzu UV-Vis 1900i Spectrophotometer in the lab as per experimental protocols.
  - **Data Analysis and Graphing**: Generated smoothed continuous spectrum plots of the data points using OriginPro2016 for post-experimental analysis. View Report*

Co-Curricular Involvement

- **Teaching Assistant: Quantum Mechanics I (Physics 464)**: Fall 2025; assisted in course instruction under Professor Socolar, held weekly meetings to discuss problem-setting, and, graded assignments and exams; helped students to refine their understanding.
- **Duke Society of Physics Students (SPS)**: Co-President (24/25) and Event Co-ordinator (23/24); led initiatives to enhance departmental engagement, organized a Physics Research Seminar Series and an Alumni Seminar Series over two years, fostering community and networking within the Duke Physics department.
- **Cultural and Visual Documentation Project**: Conducted field research and documented over 100 temples in South India and over 15 in Cambodia; developed a multimedia poster titled "South Asia's Sacred Temples: A Dive into the Ethno-Geographical and Transcultural Religious Exchange"; produced visual materials on Dravidian–Khmer architectural exchange and developed a documentary-style trailer of the visits.View Poster* Documentary Trailer*

- **Creative Project: Mycopoetica**: Authored a poetry collection exploring fungi's role in climate change via the lens of absurdist philosophy and thematic verse; structured 10 poems to provoke critical reflection and planned multimedia extensions for a college audience; project for my freshman fall course: Performing Science: Experimentation, Collaboration, and Artistry. View Collection*
- **Duke Nepali Students Association**: Founding President; Brought together 40+ Nepali-origin students and staffs at Duke; Organized 10+ bonding & cultural events, networked with the Nepali Diaspora in NC, engaged broader Duke community with Nepali cultures. View Socials*
- **Duke in Geneva – 2024**: Studied International Business and Political Philosophy of Globalization as part of Duke's Study Away Program; visited the UN, Palais des Nations, UNHCR, ILO, WHO, and WTO to explore global institutional work; also visited CERN; traveled across many cities in Europe for field experiences in Philosophy, Politics, and Economics (PPE).
- **Hobbies**: Philosophy (especially Non-Dualism), Singing and Acoustic Guitar, Trekking, Hiking, and Cricket (Duke Intramural Leagues Cricket Championship-2025). Misc Projects*.

Honors and Awards

- **Karsh International Scholarship-2022, Duke University**: A full-ride merit-based scholarship amounting over $350,000+ for exceptional skills in interdisciplinary research and civic engagement which covers "tuition, room and board, mandatory fees, demonstrated need that exceeds those costs, and three summers of funding for research, unpaid internships, and other opportunities that will advance your academic career." (April '22) View Award*
- **PSI Start, Perimeter Institute for Theoretical Physics**: Selected for a competitive 10-week theoretical physics program covering group theory and the Dirac equation, supersymmetric quantum mechanics, tensor networks in many-body systems, and cosmic structure formation via analytical and numerical methods.(May-July '25) View Program*
- **Dean's Summer Research Fellowship (DSRF), Duke University**: Fellowship to continue theoretical work on Quantum Monte Carlo methods for the Shastry-Sutherland model over the summer. (March '25)View Award*
- **First Prize, MIT Interdisciplinary Quantum Hackathon (iQuHack)**: Challenge proposed by Quantinuum, the largest quantum computing company in the world; our work focused on simulating open and closed quantum systems through their Hamiltonians and Lindbladians. (Jan.'25)View Full Project*
- **Global Fellow-2023, Duke Int'l Student Center, Office of Global Affairs, and Duke Career Center**: "It is an experiential learning and leadership program that develops students' global competency and prepares them to be true global citizens who make positive impacts around them. The students in the program also serve as the "face" of Duke's student population to visiting international scholars and guests." (April '23) View Award*
- **Huang Fellow-2023, Duke Initiative for Science and Society**: "This highly selective program fosters a community of 20 accomplished undergraduate scholars to be trained in the sciences and grounded firmly in the liberal arts–and who will be well prepared to serve as leaders in sciences"; Fully-funded summer research + career workshops. (March '23) View Award*
- **Global Awardee at Beamline for Schools-2021, CERN**: Awarded by the European Council for Nuclear Research (CERN) for research proposal on targeted cancer treatment using Delta rays; Received Alpha Ray Spectrometer, Cloud Chambers, USB devices, and other equipment (June '21) View Award*

Research Presentations and Conferences

- **APS Global Physics Summit-2026, Combined March & April Meeting** — Denver, Colorado
  Projected Thermal Density Matrix Monte Carlo on the Shastry-Sutherland Model — *Mar. 16-20, 2026*
- **International Conference on Strongly Correlated Quantum Materials - ICSCQM** — Hvar, Croatia
  Low-energy spectrum of the Shastry-Sutherland Model on Small lattices using a new Monte Carlo — *Oct. 12-17, 2025*
- **Dept. of Physics Annual Undergrad Symposium, Duke University** — Durham, NC
  Exact Diagonalization and Monte Carlo studies of the Shastry-Sutherland Model — *April 24, 2025*
- **APS Global Physics Summit-2025, Combined March & April meeting** — Anaheim, California
  Experimental synthesis, single crystal, growth, & characterization of $NdFeO_3$ — *Mar. 16-22, 2025*
- **AAAS Annual Conference-2025, Science Shaping Tomorrow** — Boston, Massachusetts
  Attended the American Association of Advancement of Science annual conference — *Feb. 13-15, 2025*
- **South Eastern APS Annual Conference-2024** — Charlotte, NC
  Attended the annual meeting as a Duke SPS representative at UNC Charlotte — *Oct. 24-26, 2024*
- **Dept. of Physics Annual Undergrad Symposium, Duke University** — Durham, NC
  Poster presented about Spinels and Perovskites synthesis and characterization — *April 25, 2024*
- **Duke Summer Research Programs Symposium** — Durham, NC
  Poster presentation of Particle-Collider track reconstruction algorithms — *July 29, 2023*
- **Dept. of Physics Annual Poster Session, Duke University** — Durham, NC
  Poster presentation on synthesis & structural characterization of Spinel compounds — *April 18, 2023*
- **9th National Conference on Science and Technology** — Kathmandu, Nepal
  Oral presentation on mustard oils in the 'Food Security and Green Economy' symposium — *June 26-28, 2022*
- **National Young Scientist Conference** — Janakpur, Nepal
  Invited lecture on study about mycotectural engineering possibilities on Mars — *June 23-24, 2022*
- **Phoenix Space Launchpad Challenge** — London, United Kingdom
  Defended research on mycotectural possibilities on Mars before a jury of scientists — *Oct. 5, 2021*
- **Asia Pacific Conference of Young Scientists** — Sonora, Hermosillo, Mexico
  National delegate presenting research proposal on Delta Rays — *Sept. 7 - Oct. 1, 2021*
- **ANPA Global Conference** — United States
  Oral presentation on research proposal about Delta Rays — *July 16-18, 2021*

Publications

- **Karna, A.**, Wu, H. S., Chandrasekharan, S., and Kaul, R. K. *Projected Density Matrix Sampling for Lattice Hamiltonians*, 2025. URL https://arxiv.org/abs/2511.19209. arXiv:2511.19209 [cond-mat.str-el] (submitted to PRD).
- **Karna, A.** *Theoretical and Experimental Investigations of Novel Quantum Magnets* (Thesis in preparation)
- Shrestha, H.*, **Karna, A.***, et al. *Investigation of the cross-border movement of* $PM_{2.5}$ *in the Terai Belt of Nepal using a low-cost sensor network.* https://doi.org/10.1016/j.apr.2026.102920 Atmospheric Pollution Research. (*Co-first author)
- Aaliya, A., **Karna, A.**, Pokhrel, S., Dahal, et al. (in review). *Global environmental change and health outcomes: Rural Communities' perspectives from Siranchowk Rural Municipality, Gorkha District, Nepal.* Discover Social Science and Health, SpringerNature.